

Medium-term determinants and developing countries: an empirical exploration

Replication and Extensions

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1. Introduction

The current account has long stood at the centre of international macroeconomics, both as a summary statistic of a country’s external position and as a window onto the deeper forces shaping saving and investment across borders. As a matter of accounting, the current account is simply the gap between national saving and investment $CA = S - I$, yet this identity conceals a rich set of structural determinants. Understanding what drives persistent surpluses and deficits matters beyond academic interest: the sharp global imbalances of the 2000s, with the United States running deficits of 5 to 6 percent of GDP while emerging Asia accumulated large surpluses, raised fundamental questions about sustainability and adjustment risks.

The intertemporal approach to the current account provides the theoretical foundation for empirical work in this area. In this framework, countries borrow against anticipated future income and run surpluses when facing temporarily high output, while fiscal expansions can spill over into external deficits through the twin deficit channel, though only partially, given the incomplete Ricardian offset documented in the data.

CP translate these predictions into an empirical framework, asking which structural fundamentals explain medium-term current account positions across a broad panel of industrial and developing countries over 1971–1995. Using five-year non-overlapping averages to filter out cyclical variation, they find that the initial net foreign asset position, the fiscal balance, demographic structure, and financial depth are significant determinants of the current account, with notable differences between industrial and developing countries.

This paper replicates their analysis over the original period (1971–1995) and extends it to a more recent sample. Following **forbeswarnock2012**, it further augments the framework by using gross capital flow measures rather than net flows, and examines how financial crises affect current account dynamics.

2. Presentation of the Paper

2.1. Empirical Framework

Motivated by the intertemporal approach to current account imbalances, **CP2003** employ a regression framework to identify which variables affect the current account and to assess whether the results are consistent with the theory. The empirical specification is a reduced-form panel model in which the current account balance, expressed as a share of GDP, is regressed on a set of structural determinants:

$$\begin{aligned} \left(\frac{CA}{GDP} \right)_{i,t} = & \alpha + \beta_1 NFA_{i,t} + \beta_2 GOVB_{i,t} + \beta_3 RELY_{i,t} + \beta_4 YGRAVG_{i,t} \\ & + \beta_5 YGRSD_{i,t} + \beta_6 RELDEPY_{i,t} + \beta_7 RELDEPO_{i,t} + \beta_8 OPEN_{i,t} + \beta_9 FDEEP_{i,t} + \beta_{10} TOTSD_{i,t} \\ & + \beta_{11} LREEER_{i,t} + \delta_t + \varepsilon_{i,t}(1) \end{aligned}$$

where δ_t denotes period fixed effects and $\varepsilon_{i,t}$ is the error term. Each regressor is motivated by the intertemporal model: the initial net foreign asset position captures the dynamics of external adjustment; the fiscal balance tests the ricardian equivalence and the twin deficit hypothesis; demographic variables proxy for life-cycle saving behaviour; trade openness and financial depth reflect structural features that affect capital attractiveness; terms-of-trade volatility captures precautionary saving motives; and relative income proxies the stage of economic development. A full description of each variable is provided in Appendix [B](#).

2.2. Data and Sample

The paper draws on a panel covering 18 industrial and 71 developing countries over the period 1971–1995, combining data from the International Financial Statistics (IFS), the World Bank Social Indicators, the World Development Indicators (WDI), and the IMF World Economic Outlook. Net foreign asset positions are taken from the Lane–Milesi-Ferretti dataset. Annual observations are collapsed into five-year non-overlapping averages, yielding up to five observations per country over the periods 1971–75, 1976–80, 1981–85, 1986–90, and 1991–95. This averaging serves two purposes: it eliminates high-frequency cyclical variation, focusing attention on the medium-run structural determinants that the theory identifies; and it reduces measurement error, which is particularly severe for developing country data at annual frequency.

2.3. Methodological Choices

Several methodological choices are worth highlighting, as they shape the interpretation of the results and directly inform the replication strategy adopted in this paper.

First, Chinn and Prasad deliberately exclude country fixed effects. Their reasoning is that approximately 40 percent of the total variation in current account balances is cross-sectional, that is, it reflects persistent differences in structural characteristics across countries. Including fixed effects would absorb this variation and prevent identification of the cross-country patterns that are the primary object of interest. Instead, period dummies are included in all panel specifications to control for time-varying global factors such as the Latin American debt crisis of the early 1980s and the surge in capital flows to emerging markets in the 1990s.

Second, the regressions are run separately for industrial and developing countries. This choice is motivated empirically: the coefficients differ significantly across groups when formally tested, suggesting that the determinants of the current account operate through different mechanisms depending on a country’s level of development. An oil-exporter dummy is included in all panel regressions to account for the structurally different saving behaviour of commodity exporters.

Third, the paper runs two distinct empirical exercises, a pure cross-section using long-run country averages over the full 1971–1995 period, and a panel using the five-year averages described above. These two exercises exploit different sources of variation and, as discussed in Section 4.2, yield somewhat different conclusions, particularly regarding the role of the initial net foreign asset position and the fiscal balance.

2.4. Key Findings

The results confirm that structural fundamentals explain a meaningful share of medium-term current account variation, with the panel $\bar{R}^2$ reaching approximately 0.44 for developing countries. The initial net foreign asset position enters positively and significantly across most specifications: countries with larger net foreign asset positions tend to run larger current account surpluses, reflecting both the investment income that creditor countries earn on their external wealth and a capital market selection effect whereby high-productivity developing countries attract persistent inflows despite accumulated liabilities. The fiscal balance is a significant determinant for developing countries, with a coefficient of approximately 0.40, implying a partial but incomplete Ricardian offset of around 60 percent; for industrial countries, the coefficient is near zero and statistically insignificant. Old-age dependency ratios enter negatively for industrial countries, consistent with life-cycle saving behaviour, but are less robust for developing countries. Financial deepening, terms-of-trade volatility, and trade openness are significant for developing countries but not for industrial countries, while GDP growth is positively and significantly associated with the current account only for industrial countries, reflecting productivity-driven saving rather than capital attraction.

3. Data Preparation

3.1. Issues Encountered

In order to run panel regressions that yield efficient estimations, a large enough sample size is required. This relies on getting enough observations where no variable used in our regression is missing, i.e., shown as a NA, over several country and years. Databases used by the original paper have been discontinued and changed over the years, leading to complications in sourcing data. Relying on the original sources did not yield enough observations, leading us to look for alternative sources not used by the original paper. Most standard database handled by the IMF or World Bank start in the early 1980's, while the paper's coverage extends to the 70's. In addition to omissions, some of the choices made in handling data had to be left to us due to a lack of clarity in the interpretation of the definition of some variable. As an example, the use of a dummy for oil exporting countries was required to run some regression, without any clear list or criteria being given by the authors. This led us to use OPEC members; without any certainty, this was the choice made in the original paper.

3.2. Data sourcing for replication

Most of the core data is sourced from the World Development Indicators (WDI) database. It was downloaded using the API package for R "WDI". This includes :

- CAGDP
- GDP per capita data, which was used to compute RELY by dividing by the US' value and setting a maximum of 1.
- RELDEPY, which was later normalized to correspond to the original paper's datapoint
- RELDEPY, which was later normalized to correspond to the original paper's datapoint
- GDP series, which were used to compute YGRAVF and YGRSD
- Terms of trade data, that was used to compute TOTSD.
- Real effective exchange rate data, that was used to compute LREEER.
- OPEN
- FDEEP
- NSGDP

Both capital controls used in the original paper were substituted for a single capital control variable, sourced from the Chinn-Ito index dataset, published by the aforementioned researchers. Data on NFAGDP¹ was obtained through data on International Investment Positions (IIP) found in the International Financial Statistics (IFS) dataset, handled by the IMF, as well as GOVBGDP. Better data on Real effective exchange rate data was sourced from Bruegel, and used to re-compute LREEER. Better data on public finances was sourced from the "Public Finances in modern history" database, hosted by the IMF. Gross expenses to GDP were subtracted from gross revenue to GDP to get a new datapoint for GOVBGDP. Data on both constant and current value of exports was downloaded from WDI, in order to compute implicit deflators and get a new datapoint for Terms of Trade, and TOTSD. Better data on GDP per capita was sourced from the Penn World Tables, and later was used to compute RELY based on US data. Data on national savings to GDP was extracted from the World Economic Outlook (WEO) database, to get earlier datapoints for NSGDP.

¹Data from IFS excludes gold from the international investment position.

3.3. New data sources for extensions

Data on gross capital flows was sourced from the database used by Broner et al. (2013)@BRONER2013, hosted on the World Bank’s website. Data on country affected by crisis (namely, Asian crisis) was arbitrarily set based on consensus given by sources on dates. All countries were marked as being affected by the great financial crisis (GFC).

3.4. Combining data sources to allow for sufficient sample size

When a single dataset was not sufficient to achieve a number of observations in the same range as the one found in the paper, we decided to merge two data sources, using the following rules :

1. Use only one data source for each country, in order to keep coherent timeseries.
2. Keep the source that had the least NA’s for the 1971-1995 period, as replication was our priority.

This was used for National Savings on GDP, where WDI and WEO data was combined; Government Balance on GDP, where two IMF datasets were combined, and the standard deviation of terms of trade, where direct WDI data was combined with implicit terms of trade derived from the gap between constant and current USD values of exports and imports, also sourced from WDI.

4. Replication of the Results

4.1. Workflow

Following in the footsteps of Adam Smith, our workflow was based on the division of labor. One person was given the task of steadily improving our datasets and implementing a self-sufficient data preparation code, while the two others were tasked with preparing code for the regression, running it on their own in order to interpret the results and check for possible data needs. This relied on git usage. Bug fixing and checking for errors was handled by all team members. Extensions were handled by all members, albeit using different codes in order to avoid git conflicts. AI was used routinely, namely Claude, to automate tedious coding tasks (properly formatting tables with stargazer, building small code snippets to bridge gaps in R knowledge, handling libraries we had not encountered before). Claude was also helpful in finding some data sources we had not encountered in our initial research, allowing us to bridge data gaps and run a proper replication. All tasks related to interpretation, reflection, organization and quality control were otherwise done by group members.

4.2. Comparison with Original Results

The variance decomposition in Table 5 reveals an initial divergence from the original. **CP** find that for industrial countries, approximately 39% of current account variation is cross-sectional and 61% is over time, with a broadly similar split for developing countries. Our replication reverses this pattern for industrial countries, 52% of variation is cross-sectional and 48% over time, while finding more time-series variation for developing countries (67%) than the original (56%). These differences likely reflect changes in data vintages and sample coverage, but the methodological implication is worthy: our decomposition actually strengthens the original authors’ justification for preferring OLS over fixed effects in the industrial country sample, since cross-sectional variation is even more prominent in our data.

The most consequential divergence concerns the fiscal balance result. The headline finding of **CP** a robust positive association between the government budget balance and the current account, with a coefficient of approximately 0.40 for developing countries in the panel, does not

survive in our replication. Across the cross-section regressions (Tables 6 and 7), the panel OLS specification (Table 8), and the fixed effects specifications (Tables 9 and 10), the fiscal balance coefficient is small, unstable, and statistically insignificant throughout. By contrast, the initial NFA position replicates robustly across all specifications: after accounting for a scaling difference in the NFA/GDP variable, our panel coefficient of 4.464 reported in Table 8 is consistent with the original’s 0.048, and the result remains significant for both industrial and developing countries. Current account persistence in the annual regressions (Table 11) also replicates closely, with lagged CA coefficients of 0.55–0.59 in our data against 0.57 in the original.

Two additional results in our replication differ from the original in theoretically interesting ways. First, Table 8 reveals a significant negative relationship between GDP growth and the current account in the panel (-0.245^{**} for the full sample), a result **Chinn2003** do not find for developing countries. This is consistent with the interpretation that higher growth attracts capital inflows and finances larger current account deficits, a mechanism that may be more prominent in the post-1995 period if our sample extends beyond the original’s endpoint. Second, the lagged real effective exchange rate enters significantly and negatively in our annual regressions reported in Table 11 (-4.011^{***}), while the original finds this effect negative but insignificant. Demographic variables also emerge as more significant in our panel (Table 8) than in the original, with both youth and old dependency ratios reaching conventional significance levels. Taken together, these differences suggest that while the structural role of NFA positions is a durable empirical regularity, the fiscal balance result is more sensitive to data vintage, sample composition, and time period, a fragility worth flagging given its central role in the original paper’s policy interpretation.

5. Extensions

5.1. Actualization of the Sample

After the actualization of the sample until today with five-years data, we can re estimate precedent coefficients and results with more data and updated data.

Before turning to the regression results, the table 15 documents the between and within-country variance decomposition for the key variables in the updated sample.

The broad structure of the variance decomposition is preserved: variables reflecting structural country characteristics: relative income, trade openness and financial deepening remain predominantly cross-sectional, while variables capturing macroeconomic dynamics like average GDP growth, terms of trade volatility and the real exchange rate are driven primarily by within-country variation over time. This pattern closely mirrors the original.

Several notable differences emerge. The current account ratio exhibits a roughly equal split between between- and within-country variation for developing countries (44%/56%), contrary to the original decomposition. The NFA position, as expected, is predominantly cross-sectional (44% between for industrial, 44% for developing), though its between share is lower than in the original paper (80% and 67%), reflecting the substantial realignment of external positions generated by two decades of global imbalances and crisis episodes.

Graphs from 3 and 4 in appendix which reproduce graphs of the paper are close of the paper for the most of them contrary to graphs edited with the period 1971-1995.

The updated cross-section estimates (table 16) confirm the two most robust findings of Chinn and Prasad: the government budget balance and the initial NFA position are positively and significantly associated with the current account across all country groups. On the full sample, a one-percentage-point improvement in the fiscal balance is associated with a 0.359 percentage point improvement in the current account, close to the original estimate of 0.375. The NFA coefficient (1.427, $p < 0.001$) is substantially larger than in the original paper (0.046), reflecting

the extended time span and the broader variation in external positions generated by two decades of additional data. Financial deepening enters negatively and marginally significantly for the full and developing samples, in contrast to the positive sign found by Chinn and Prasad for developing countries, a result we revisit in the Ricardian offset extension below. Trade openness and the oil exporter dummy retain their expected signs and significance. The dependency ratios are significant for the full and developing samples, consistent with the life-cycle saving channel, but lose significance once Africa is excluded, suggesting that the result is partly driven by the demographic structure of low-income African economies.

The five-specification panel OLS replication (table 1) yields a fiscal coefficient of 0.323 for the full sample ($p < 0.01$), slightly above the original 0.306, and rising to 0.348 for developing countries excluding Africa. These estimates are all statistically significant and remarkably stable across the three non-industrial samples, suggesting that the twin-deficit relationship has, if anything, strengthened over the post-1995 period. For industrial countries, the coefficient (0.447, $p < 0.01$) is now significant—unlike in the original paper—which may reflect the sharper fiscal consolidations and expansions associated with the Global Financial Crisis and its aftermath. The NFA position is positive but imprecisely estimated across all specifications, consistent with the original finding that this variable captures primarily cross-sectional rather than within-country variation. Among the controls, capital account openness is negative and significant for non-industrial groups, average GDP growth is negatively associated with the current account, and trade openness retains a positive and significant coefficient for all but the developing country full sample—a sign reversal relative to the original paper that likely reflects the stronger integration of emerging markets into global value chains over the updated period.

The within-estimator results of table 17 provide a demanding robustness check by absorbing all time-invariant country characteristics. The fiscal coefficient remains positive and highly significant across all five samples, ranging from 0.265 (developing excluding Africa) to 0.827 (industrial countries). The industrial country estimate is particularly large and precisely estimated, in contrast to the original paper where the FE coefficient was 0.141 and insignificant. This strengthening is consistent with the post-2008 fiscal dynamics of advanced economies, where large and heterogeneous fiscal consolidations provide within-country variation that was largely absent in the 1971–1995 sample. The NFA coefficient shrinks considerably once country fixed effects are included, as expected given that NFA positions are highly persistent and thus identified primarily in the cross section. Relative income terms gain significance in the within-dimension for several samples and display the sign pattern consistent with the stages-of-development hypothesis, negative level, positive quadratic as Chinn and Prasad also found under FE estimation. Overall, the Hausman-style comparison between OLS and FE results suggests that country fixed effects matter for NFA and relative income, but that the fiscal balance result is robust to specification choice.

The annual specification (table 18) introduces two important additions: a lagged current account ratio and a lagged change in the real effective exchange rate. The estimated persistence coefficient ranges from 0.686 to 0.762 across country groups, indicating substantial but incomplete inertia in current account positions at annual frequencies. The fiscal coefficient falls relative to the five-year averaged specifications from 0.323 to 0.197 for the full sample which is expected, since the lagged dependent variable absorbs part of the dynamic adjustment and the higher-frequency variation in fiscal balances contains more cyclical noise. The relative price effect is negative and significant across all samples: a lagged appreciation of the real effective exchange rate of one percent is associated with a deterioration from 2.5 to 7.9 percentage points in the current account, with the largest effect estimated for industrial countries. Financial deepening enters positively and significantly at annual frequencies for the full and developing samples, in contrast to its negative sign in the five-year specifications, suggesting that the negative medium-term relationship reflects a structural association between financial depth and investment rather than

Table 1: OLS with time effects.

	(1) Full	(2) Full excl. Africa	(3) Industrial	(4) Developing	(5) Dev. excl. Africa
Govt. budget balance	0.323*** (0.077)	0.309*** (0.064)	0.447*** (0.159)	0.344*** (0.088)	0.348*** (0.064)
NFA to GDP	1.469 (1.084)	0.934 (0.915)	1.777 (1.409)	0.746 (0.896)	0.095 (0.491)
Relative income	0.196 (4.608)	-10.752*** (4.067)	55.625 (49.552)	-0.795 (5.617)	-15.502*** (5.066)
Relative income ²	3.704 (4.244)	13.652*** (3.811)	-27.300 (30.208)	14.491** (6.919)	29.893*** (6.411)
Rel. dependency (young)	-3.541*** (0.921)	-0.178 (1.217)	0.146 (12.137)	-4.107*** (0.973)	-0.345 (1.167)
Rel. dependency (old)	-1.606** (0.718)	-0.031 (0.676)	1.779 (1.728)	-3.140*** (0.935)	-0.339 (0.768)
Financial deepening	-0.001* (0.0003)	0.016* (0.008)	0.016 (0.016)	-0.001** (0.0003)	0.016 (0.014)
ToT volatility	0.001 (0.037)	-0.018 (0.028)	0.131 (0.211)	-0.006 (0.038)	-0.029 (0.028)
Avg. GDP growth	-0.205** (0.087)	-0.240*** (0.086)	-1.012** (0.449)	-0.168* (0.090)	-0.201** (0.088)
Openness	0.016** (0.008)	0.025*** (0.009)	0.052** (0.023)	-0.001 (0.009)	0.006 (0.012)
Capital controls	-1.296* (0.665)	-2.485*** (0.737)	3.704 (2.796)	-1.246* (0.735)	-2.462*** (0.836)
Oil exporter dummy	2.475 (1.717)	5.357*** (1.204)	—	2.498 (1.724)	5.646*** (1.077)
Constant	2.517 (1.912)	-1.019 (2.013)	-37.071 (27.556)	4.885** (2.143)	0.461 (2.035)
Time dummies	Yes	Yes	Yes	Yes	Yes
Observations	504	357	69	435	288
R^2	0.350	0.405	0.770	0.362	0.462
Adjusted R^2	0.322	0.368	0.674	0.329	0.419

Notes: Dep. var.: current account to GDP ratio. OLS with time dummies. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

a short-run behavioral channel. Capital account openness is negative and significant for non-industrial groups, consistent with the view that greater financial integration facilitates current account deficits. The R^2 of 0.66 for the full sample and 0.87 for industrial countries indicate a good overall fit, though a substantial share of the explained variance is mechanical, attributable to the lagged dependent variable.

5.2. Explaining Financial Crises

Event Study: Current Account Adjustment Around Crisis Episodes

Before estimating the regression model with crisis interactions, it is instructive to examine the raw current account dynamics around each episode non-parametrically. The event study plots the average CA/GDP ratio for each country group over a $[-5, +5]$ year-window centred on the crisis year, providing visual confirmation that the Asian crisis (1997), the Argentine crisis (2001), and the Global Financial Crisis (2008) generate sufficiently sharp and heterogeneous current account responses to warrant separate treatment in the regression.

Figure 1 documents the Asian and Argentine episodes. We see a pronounced pre-crisis deterioration in the current account, followed by a sharp reversal at $t = 0$ and a sustained improvement over the following years. For the five Asian economies (Thailand, Korea, Indonesia, Malaysia, Philippines), the average CA/GDP swings from approximately -5% in the two years preceding the crisis to nearly $+8\%$ at $t = +1$, a correction of over 13 percentage points. Argentina exhibits an almost identical profile. This J-shaped adjustment is consistent with the standard current account reversal mechanism: the sudden stop in capital inflows forces an abrupt compression of the external deficit, typically through a collapse in domestic absorption rather than an export boom. The symmetry between the two episodes is notable and justifies pooling them into a single crisis dummy.

Event study, adjustment of CA for asian and argentine's crises

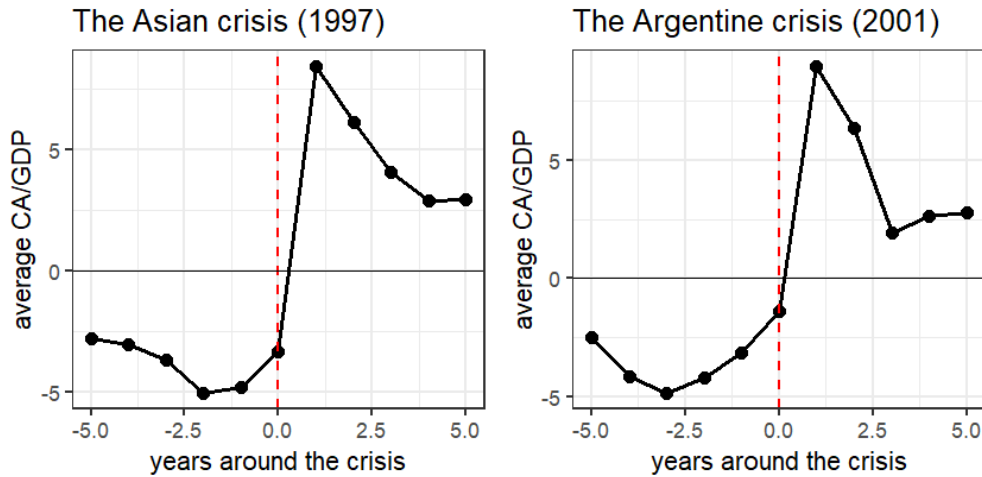


Figure 1: Event study: current account adjustment around the Asian (1997) and Argentine (2001) crises. Each point represents the average CA/GDP ratio across the relevant countries, centred on the crisis year ($t = 0$, dashed red line).

Figure 2 presents the general financial crisis of 2008 disaggregated by country group. The pattern is markedly different from the two preceding crises. All three groups record a trough at $t = 0$, but the post-crisis trajectories diverge sharply: industrial countries experience a strong and sustained current account improvement, while developing countries remain in persistent deficit, hovering around -3% throughout the post-crisis window. This heterogeneity reflects

the asymmetric nature of the GFC, a shock originating in advanced economies that triggered a fiscal and demand collapse primarily in industrial countries, whereas developing economies faced an external demand shock without the same domestic financial disruption. The divergence also provides the empirical rationale for interacting the crisis dummy with country-group indicators in the regression.

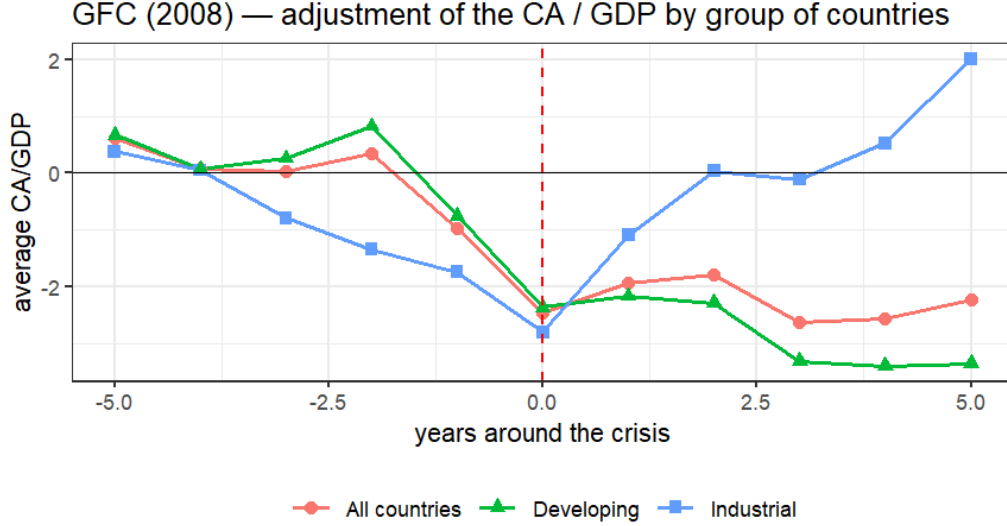


Figure 2: Event study: current account adjustment around the Global Financial Crisis (2008) by country group. The dashed red line marks the crisis year ($t = 0$).

5.3. Saving Decomposition and the Ricardian Offset

Using the accounting identity $CA = NS - I$, the difference $\beta^I = \beta^{NS} - \beta^{CA}$ captures the investment response to a fiscal shock. On the full sample ($0.393 - 0.323 = 0.070$) and excluding Africa ($0.232 - 0.309 = -0.077$), this gap is negligible in both magnitude and sign, indicating that investment plays no meaningful role in transmitting fiscal impulses to the current account. The analysis therefore focuses on the saving channel.

The benchmark regressions in Chinn and Prasad use national saving as the dependent variable without distinguishing between its public and private components, obscuring the behavioral channel through which fiscal policy affects the current account. To address this, private saving is defined as $S_{it}^{priv} = NS_{it} - GOV_{it}$ and the baseline specification is augmented with an interaction between the fiscal balance and financial deepening:

$$S_{it}^{priv} = \alpha + \beta_1 GOV_{it} + \beta_2 FDEEP_{it} + \beta_3 (GOV_{it} \times FDEEP_{it}) + \mathbf{X}'_{it}\gamma + \lambda_t + \varepsilon_{it} \quad (2)$$

A value of $\beta_1 = -1$ would indicate full Ricardian equivalence; the interaction term β_3 tests whether the offset is stronger in more financially developed economies, for which theory predicts $\beta_3 < 0$.

The following table reports the estimates. Two results stand out. First, the Ricardian offset is large and highly significant across all non-industrial groups: a one-point improvement in the fiscal balance reduces private saving by 0.625 points on the full sample, rising to 0.754 for developing countries and 1.140 excluding Africa, substantially above the offset of 0.766 implied by the original Chinn and Prasad estimates, consistent with greater macroeconomic credibility and deeper financial integration over the post-1995 period. Industrial countries show no significant offset, plausibly reflecting endogeneity of fiscal policy near potential output. Second, the interaction term $GOV \times FDEEP$ is economically negligible: evaluated at the 25th, 50th, and 75th

percentiles of the financial deepening distribution, the total marginal effect $\hat{\beta}_1 + \hat{\beta}_3 \times FDEEP$ varies from -0.749 to -0.741 , essentially flat. Financial deepening does not mediate the offset, suggesting that precautionary saving motives or informal credit networks, rather than access to formal financial markets, drive the behavioral response.

Table 2: Heterogeneous Ricardian Offset-Regression on private saving to GDP ratio.

	(1) Full	(2) Full excl. Africa	(3) Industrial	(4) Developing	(5) Dev. excl. Africa
Govt. budget balance (GOV)	-0.625*** (0.119)	-1.065*** (0.200)	0.087 (0.322)	-0.754*** (0.117)	-1.140*** (0.195)
Financial deepening (FDEEP)	-0.0004 (0.0002)	-0.003 (0.023)	-0.065** (0.026)	-0.0003 (0.0003)	0.080*** (0.022)
GOV $\times$ FDEEP	0.0003 (0.0002)	0.006 (0.004)	-0.007 (0.005)	0.0002 (0.0002)	0.006* (0.003)
Time dummies	Yes	Yes	Yes	Yes	Yes
Observations	521	367	76	445	291
R^2	0.321	0.348	0.881	0.377	0.434
Adjusted R^2	0.291	0.307	0.834	0.344	0.388

Notes: Dep. var.: private saving to GDP ratio. Panel estimation with time dummies. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.4. Gross Flow analysis

This extension is inspired by Forbes and Warnock (2012), who pioneered the analysis of gross flows during crisis. Gross flows, i.e total capital inflows and outflows, are interesting to delve into as they may reveal a reality that current account analysis may fail to notice. A country can see massive decrease in both outflows and inflows, but still see negligible changes in current account. We tried to see if the results they obtained were still seen in our extended dataset. For this, we ran time fixed-effects panel regressions with total gross flows to GDP (outflows + inflows) as our dependent variable, and keeping our list of controls. Robust standard errors were used to compute variance (Table 10). Trends in gross flows aren't robustly linked to any domestic tendency, except relative income and the openness ratio. A higher NFA seems to greatly reduce gross flows in industrial countries only, which seems to indicate that country that hold more wealth abroad see less flows than other. This may be due to these economies less often having wealth crossing the border, due to economies having to use trade and financial movement to compensate for a commercial deficit or surplus. We also ran regression to explain both inflows (Table 11) and outflows (Table 12), to similar results for outflows. However, inflows are strongly hindered by a higher government balance and higher growth, due to a lower financing need and a higher domestic financing capacity. However, a lack of time prevented us from replicating Forbes and Warnock's approach of detecting sudden episodes via standard deviation from a rolling mean, leaving us with only these structural tendencies.

We also find evidence of higher capital controls being robustly linked with a higher inflows/GDP ratio. Our hypothesis was that this was due to governments wanting to limit already high inflows by increasing controls, that we tested by adding a lagged variable (Table 13). All estimated coefficients of previous period capital controls are negative, but are not significant enough to reject the null hypothesis. However, they still show a trend of being negative, seeming to indicate that capital controls tend to rise alongside flows.

5.5. Logit approach to financial crises

We used a time fixed-effect logit approach on the asian crisis (Table 4), as it only applied to some countries and not other. Our goal was to see what structural and temporary elements that heighten the probability of the binary event, but also allows to look at some of the specific dynamics happening before the financial crisis. For this, we created a dummy variable for the

Table 3: Gross Flows OLS with time effects

	<i>Dependent variable:</i>				
	Gross Flows				
	Full	Full excl. Africa	Industrial	Developing	Dev. excl. Africa
	(1)	(2)	(3)	(4)	(5)
Govt. budget balance	0.175 (0.203)	0.242 (0.236)	0.138 (0.789)	0.342 (0.274)	0.370 (0.335)
NFA to GDP ratio	-0.691 (3.636)	-1.374 (3.984)	-31.189*** (6.614)	2.494*** (0.922)	2.254*** (0.781)
Relative income	-6.188 (15.155)	-12.278 (14.887)	-136.377 (223.716)	-27.323** (11.071)	-31.379*** (11.679)
Relative income squared	35.691*** (13.771)	40.060*** (13.804)	139.903 (136.549)	66.153*** (13.573)	68.026*** (13.925)
Rel. dependency (young)	-2.623 (4.486)	-2.163 (4.505)	2.064 (77.660)	-0.743 (4.951)	-0.140 (4.693)
Rel. dependency (old)	-3.263 (3.761)	-1.543 (3.445)	0.909 (9.361)	1.958 (4.413)	3.294 (4.191)
Financial deepening	0.004 (0.039)	-0.002 (0.042)	0.192* (0.116)	0.043 (0.058)	0.035 (0.066)
ToT volatility	0.002 (0.164)	0.019 (0.161)	1.158** (0.521)	-0.124 (0.155)	-0.110 (0.154)
Avg. GDP growth	0.503 (0.506)	0.501 (0.536)	-0.888 (2.382)	0.561 (0.477)	0.558 (0.510)
Openness ratio	0.115*** (0.029)	0.131*** (0.030)	0.531*** (0.185)	0.062* (0.034)	0.075** (0.037)
Capital controls	-4.015 (6.037)	-5.802 (6.728)	1.851 (10.394)	-0.075 (6.412)	-1.128 (7.025)
Oil exporter dummy	-1.281 (4.414)	3.636 (4.202)		1.069 (3.954)	3.407 (4.093)
Constant	-1.976 (6.751)	-3.219 (6.730)	-26.113 (139.673)	3.280 (6.633)	2.010 (6.099)
Time dummies	Yes	Yes	Yes	Yes	Yes
Observations	295	266	65	230	201
R ²	0.354	0.367	0.698	0.473	0.486
Adjusted R ²	0.309	0.318	0.580	0.426	0.432

Note:

*p<0.1; **p<0.05; ***p<0.01

three years preceding the crisis, to see what specifics the country that ended up getting hit had. We show that they had significantly higher inflows, lower relative income, a higher government balance and higher savings than other countries who weren't hit. Capital controls tended to be slightly higher in these countries, and financial deepening lower. This is a surprising result, that defies common beliefs about crisis risk factor.

Table 4: Logit year-fixed effect model of pre-asian crisis years

Dependent Variable:	abouttoasiancrisis
Model:	(1)
<i>Variables</i>	
Inflows/GDP	0.5543*** (0.1926)
Outflows/GDP	0.0281 (0.0693)
NFA/GDP	1.539 (1.437)
Relative income	-16.54** (6.940)
Fiscal balance	0.9558** (0.3835)
National Savings/GDP	0.4497** (0.1764)
Capital controls	3.951* (2.154)
Financial Deepening	-0.0611* (0.0346)
<i>Fixed-effects</i>	
year	Yes
<i>Fit statistics</i>	
Observations	131
Squared Correlation	0.61425
Pseudo R ²	0.67092
BIC	81.511
<i>IID standard-errors in parentheses</i>	
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>	

6. Conclusion

7. Appendix

A. Critics of imputation

While data imputation is frequently employed to address missing values, its practical utility in this specific framework remains highly questionable and ultimately redundant. Replacing missing entries with statistical estimates often introduces severe bias, artificially inflates statistical power, and systematically distorts the true underlying variance of the dataset. Furthermore, this approach masks data quality deficiencies rather than resolving them, while failing to provide any genuine incremental value compared to analyzing the fully observed sample. Consequently,

relying on such techniques introduces critical methodological flaws without offering any robust analytical advantage.

B. Variable Descriptions

This appendix provides a detailed description of all variables entering the baseline regression eq:cp, following the notation of **CP**.

NFAGDP — Initial net foreign asset position, expressed as a ratio to GDP. Measured at the beginning of each five-year period. Source: Lane–Milesi-Ferretti dataset. This variable captures the stock of external wealth accumulated by a country and proxies for the intertemporal budget constraint: net debtors must eventually run trade surpluses to service their liabilities.

GOVBGDP — Government fiscal balance as a share of GDP. The inclusion of this variable tests the twin deficit hypothesis: a deterioration in the fiscal balance reduces national saving and, if the Ricardian offset is incomplete, reduces the current account. Source: IMF WEO.

RELY — Relative income, measured as real GDP per capita relative to the United States. This variable proxies the stage of economic development and tests the stages-of-development hypothesis, which predicts a U-shaped relationship between income and the current account. Source: Penn World Tables.

YGRAVG — Average real GDP growth rate over the five-year period. Higher growth may attract foreign capital in developing countries (negative effect on the CA) or reflect productivity-driven saving in industrial countries (positive effect). Source: World Development Indicators.

YGRSD — Standard deviation of real GDP growth over the five-year period. This variable captures income volatility and proxies for precautionary saving motives. Source: World Development Indicators.

RELDEPY — Youth dependency ratio, defined as the population under 15 relative to the population aged 15 to 65. A higher youth dependency ratio reduces household saving, consistent with life-cycle theory. Source: World Bank Social Indicators.

RELDEPO — Old-age dependency ratio, defined as the population over 65 relative to the population aged 15 to 65. A higher old-age dependency ratio is associated with dissaving in retirement, reducing the current account. Source: World Bank Social Indicators.

OPEN — Trade openness, measured as the ratio of exports plus imports of goods and non-factor services to GDP. More open economies may attract greater foreign capital inflows, widening the current account deficit, particularly in developing countries. Source: World Development Indicators.

FDEEP — Financial deepening, measured as the ratio of M2 to GDP. A deeper domestic financial sector reduces the need for external financing and may be associated with higher domestic saving, improving the current account in developing countries. Source: International Financial Statistics.

TOTSD — Terms-of-trade volatility, measured as the standard deviation of the terms of trade over the five-year period. Greater volatility raises precautionary saving, improving the current account, particularly in commodity-dependent developing countries. Source: IFS.

LREEER — Logarithm of the trade-weighted real effective exchange rate. A real appreciation reduces export competitiveness and tends to worsen the trade balance, all else equal. Source: IFS.

NSGDP — National saving measured as a share of GDP, capturing the portion of national income that is not consumed by households, firms, or the government. Higher savings rates are generally associated with stronger external balances and greater domestic financing capacity. Source: World Development Indicators.

C. Replication results

Variable	Across (Industrial)	Over time (Industrial)	Across (Developing)	Over time (Developing)
CAGDP	51.94	48.06	32.88	67.12
CombinedGOVBGDP	66.13	33.87	64.70	35.30
PennRELY	95.18	4.82	90.58	9.42
PennRELY2	94.80	5.20	83.40	16.60
RELDEPY	76.45	23.55	85.39	14.61
RELDEPO	84.29	15.71	94.91	5.09
YGRAVG	21.00	79.00	37.85	62.15
YGRSD	22.90	77.10	35.16	64.84
CombinedTOTSD	28.79	71.21	30.67	69.33
BRREER	51.08	48.92	60.44	39.56
OPEN	95.04	4.96	94.39	5.61
FDEEP	89.46	10.54	81.18	18.82
COMBINEDNSGDP	79.75	20.25	71.84	28.16
ka_open	67.15	32.85	75.12	24.88
NFAGDP	88.53	11.47	63.12	36.88

Table 5: Decomposition of variance into cross-section and time-series components (in percent)

D. Extension results

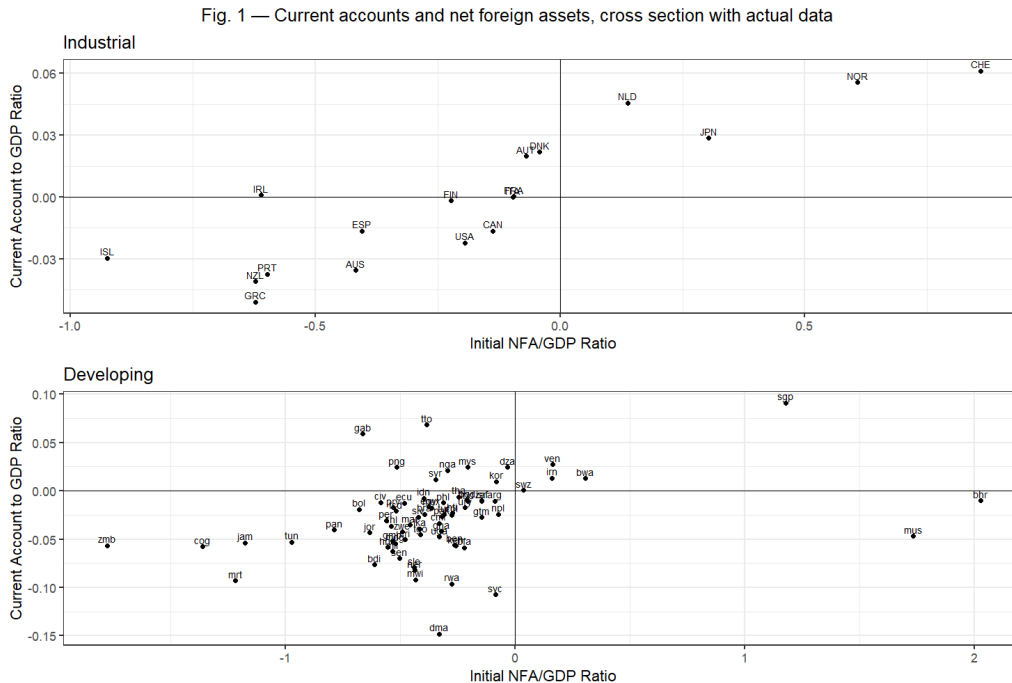


Figure 3: highlights a clear positive correlation between initial NFA/GDP and current account ratios for industrial nations, where positive asset positions yield systematic surpluses. Conversely, developing countries are heavily clustered in negative NFA territory, demonstrating a highly fragmented current account dispersion around zero.

Table 6: Cross-section regressions (current account to GDP ratio)

	<i>Dependent variable:</i>		
	Current account to GDP ratio		
	Full sample	Full excl. Africa	Industrial
	(1)	(2)	(3)
Govt. budget balance	0.075 (0.072)	0.050 (0.065)	−0.130 (0.325)
NFA to GDP ratio	3.947*** (1.384)	2.025 (1.271)	10.495 (8.963)
Relative income	5.908 (4.157)	−3.694 (4.549)	−23.835 (46.044)
Relative income squared	−4.161 (4.149)	4.916 (4.437)	21.715 (29.817)
Rel. dependency (young)	−1.597 (1.234)	−1.245 (1.425)	10.483 (19.234)
Rel. dependency (old)	−1.161 (0.953)	−0.255 (1.055)	4.636 (3.065)
GDP growth	−0.196** (0.089)	−0.181** (0.090)	−0.364 (0.653)
GDP growth volatility	0.034 (0.130)	0.070 (0.157)	1.150 (0.788)
Terms of trade volatility	0.046* (0.026)	0.012 (0.026)	0.074 (0.181)
Real exchange rate	−0.069 (0.443)	−0.186 (0.499)	−4.103 (7.727)
Trade openness	0.009 (0.011)	0.022* (0.011)	0.138** (0.055)
Financial deepening	0.033 (0.020)	0.009 (0.021)	−0.137* (0.079)
Capital account openness	−1.699 (1.155)	−1.241 (1.130)	11.501** (5.209)
Oil exporter dummy		3.177* (1.801)	
Observations	172	141	26
R ²	0.531	0.520	0.683
Adjusted R ²	0.493	0.467	0.367

Note:

Table 7: Cross-section regressions (developing countries)

	<i>Dependent variable:</i>	
	Current account to GDP ratio Developing	Dev. excl. Africa
	(1)	(2)
Govt. budget balance	0.106 (0.104)	0.151 (0.104)
NFA to GDP ratio	3.737** (1.456)	1.410 (1.276)
Relative income	4.424 (8.066)	-12.445 (9.165)
Relative income squared	0.818 (13.972)	24.000 (15.785)
Rel. dependency (young)	-1.571 (1.279)	-0.977 (1.387)
Rel. dependency (old)	-1.396 (1.294)	-0.105 (1.272)
GDP growth	-0.200** (0.095)	-0.197** (0.092)
GDP growth volatility	0.031 (0.145)	0.074 (0.161)
Terms of trade volatility	0.051 (0.032)	0.011 (0.027)
Real exchange rate	-0.011 (0.515)	-0.045 (0.540)
Trade openness	0.007 (0.012)	0.020* (0.010)
Financial deepening	0.034 (0.023)	0.005 (0.023)
Capital account openness	-1.681 (1.382)	-1.444 (1.354)
Oil exporter dummy	-0.328 (2.290)	3.583** (1.789)
Observations	146	115
R ²	0.537	0.541
Adjusted R ²	0.488	0.477

*Note:*¹⁹*p<0.1; **p<0.05; ***p<0.01

Table 8: OLS with time effects

	<i>Dependent variable:</i>				
	Current Account to GDP ratio				
	Full	Full excl. Africa	Industrial	Developing	Dev. excl. Africa
	(1)	(2)	(3)	(4)	(5)
Govt. budget balance	0.034 (0.077)	0.008 (0.072)	0.300 (0.213)	0.035 (0.107)	0.082 (0.109)
NFA to GDP ratio	4.464*** (1.469)	2.471** (1.210)	7.654* (4.444)	4.264*** (1.494)	1.881 (1.215)
Relative income	5.935 (5.488)	−4.038 (5.329)	320.472** (154.601)	3.968 (9.999)	−12.904 (9.714)
Relative income squared	−3.525 (5.259)	5.884 (5.009)	−170.225* (89.283)	2.122 (16.431)	24.929 (16.070)
Rel. dependency (young)	−3.256* (1.680)	−2.523 (1.797)	−45.668** (18.714)	−3.687* (1.974)	−2.876 (1.765)
Rel. dependency (old)	−2.094** (1.059)	−1.044 (1.042)	−0.995 (4.074)	−3.098** (1.573)	−1.704 (1.327)
Financial deepening	0.021 (0.020)	−0.003 (0.021)	−0.149* (0.087)	0.021 (0.022)	−0.006 (0.022)
ToT volatility	0.055 (0.034)	0.012 (0.029)	1.040*** (0.265)	0.051 (0.035)	0.007 (0.030)
Avg. GDP growth	−0.245** (0.096)	−0.241*** (0.086)	−0.168 (0.426)	−0.254** (0.104)	−0.262*** (0.091)
Openness ratio	0.008 (0.011)	0.022** (0.011)	0.111** (0.048)	0.004 (0.012)	0.017* (0.010)
Capital controls	−1.953 (1.237)	−1.543 (1.171)	3.953 (5.443)	−1.463 (1.423)	−1.307 (1.357)
Oil exporter dummy	−0.463 (2.169)	3.195** (1.562)		−0.538 (2.205)	3.579** (1.576)
Constant	2.145 (2.999)	1.557 (2.881)	−128.828* (73.123)	3.599 (4.140)	3.429 (3.300)
Time dummies	Yes	Yes	Yes	Yes	Yes
Observations	172	141	26	146	115
R ²	0.310	0.303	0.822	0.323	0.347
Adjusted R ²	0.239	0.213	0.556	0.239	0.240

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 9: Fixed effects panel regressions

	<i>Dependent variable:</i>		
	Current account to GDP ratio		
	Full sample	Full excl. Africa	Industrial
	(1)	(2)	(3)
Govt. budget balance	−0.059 (0.104)	−0.093 (0.112)	0.035 (0.383)
NFA to GDP ratio	1.088** (0.465)	0.635*** (0.190)	15.781*** (5.937)
Relative income	29.862 (20.823)	11.923 (18.882)	343.502** (170.899)
Relative income squared	−11.540 (23.747)	7.481 (22.664)	−184.154** (92.494)
Rel. dependency (young)	3.339 (4.208)	1.147 (3.799)	5.378 (61.443)
Rel. dependency (old)	−1.167 (6.356)	3.903 (5.389)	18.849 (23.645)
Financial deepening	−0.014 (0.033)	0.016 (0.029)	−0.008 (0.207)
Capital account openness	0.003 (2.054)	0.861 (2.165)	19.374* (10.025)
Observations	197	156	26
R ²	0.083	0.092	0.570
Adjusted R ²	−0.438	−0.394	−0.343

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 10: Fixed effects panel regressions (developing countries)

	<i>Dependent variable:</i>	
	Current account to GDP ratio	
	Developing	Dev. excl. Africa
	(1)	(2)
Govt. budget balance	−0.049 (0.116)	−0.078 (0.122)
NFA to GDP ratio	1.037** (0.476)	0.527** (0.215)
Relative income	39.314 (31.470)	4.744 (26.167)
Relative income squared	−27.077 (48.179)	22.132 (40.023)
Rel. dependency (young)	3.304 (4.588)	0.025 (3.961)
Rel. dependency (old)	−1.583 (7.718)	6.850 (5.795)
Financial deepening	−0.015 (0.033)	0.010 (0.029)
Capital account openness	−0.783 (2.114)	0.360 (2.194)
Observations	171	130
R ²	0.087	0.111
Adjusted R ²	−0.477	−0.415
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01	

Table 11: OLS annual data between 1971 and 1995 with time effects

	<i>Dependent variable:</i>				
	Current Account to GDP ratio				
	Full	Full excl. Africa	Industrial	Developing	Dev. excl. Africa
	(1)	(2)	(3)	(4)	(5)
Govt. budget balance	0.071* (0.038)	0.043 (0.037)	0.075 (0.089)	0.072 (0.048)	0.072 (0.050)
NFA to GDP ratio	1.358** (0.671)	0.745 (0.470)	9.581*** (3.366)	1.302* (0.667)	0.603 (0.499)
Relative income	8.165*** (2.691)	2.620 (2.399)	57.550 (50.045)	8.997** (4.360)	-2.833 (4.195)
Relative income squared	-7.164*** (2.422)	-2.050 (2.162)	-46.840 (29.601)	-8.706 (6.594)	8.168 (6.702)
Rel. dependency (young)	0.989 (0.993)	0.137 (0.997)	17.466 (11.070)	1.160 (1.159)	-0.495 (1.094)
Rel. dependency (old)	0.438 (0.427)	0.623 (0.432)	1.791 (1.928)	0.395 (0.599)	0.243 (0.556)
Financial deepening	0.007 (0.012)	-0.005 (0.011)	0.016 (0.031)	0.010 (0.015)	-0.009 (0.014)
ToT volatility	0.002 (0.004)	0.002 (0.005)	0.094*** (0.032)	0.001 (0.004)	0.003 (0.005)
Avg. GDP growth	-0.075 (0.047)	-0.082 (0.051)	-0.005 (0.129)	-0.068 (0.050)	-0.083 (0.053)
Openness ratio	-0.019** (0.009)	-0.011 (0.007)	0.014 (0.026)	-0.023* (0.012)	-0.013 (0.009)
Capital controls	-0.507 (0.479)	-0.602 (0.454)	3.489* (2.082)	-0.423 (0.642)	-0.532 (0.565)
Oil exporter dummy	-0.294 (1.197)	1.223 (1.097)		-0.323 (1.217)	1.486 (1.094)
Lagged CA/GDP	0.594*** (0.047)	0.545*** (0.050)	0.450** (0.181)	0.577*** (0.050)	0.515*** (0.053)
Lagged log REER	-4.011*** (0.824)	-3.420*** (0.763)	-3.023 (3.417)	-3.882*** (0.856)	-3.187*** (0.767)
Constant	-0.574 (1.944)	1.250 (1.873)	-40.232** (19.742)	-0.362 (2.583)	4.009* (2.344)
Year dummies	Yes	Yes	Yes	Yes	Yes
Observations	612	516	111	501	405
R ²	0.539	0.516	0.740	0.536	0.523
Adjusted R ²	0.510	0.480	0.619	0.500	0.476

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 12: Gross Inflows OLS with time effects

	<i>Dependent variable:</i>				
	Full	Full excl. Africa	Inflows Industrial	Developing	Dev. excl. Africa
	(1)	(2)	(3)	(4)	(5)
Govt. budget balance	−0.200** (0.092)	−0.236*** (0.072)	−0.398*** (0.109)	−0.165 (0.111)	−0.220*** (0.077)
NFA to GDP ratio	−0.785 (0.895)	−0.803 (0.991)	−5.170*** (0.625)	0.067 (0.336)	0.120 (0.335)
Relative income	7.814* (4.721)	8.503* (4.440)	0.093 (29.491)	0.896 (4.732)	1.279 (4.466)
Relative income squared	−8.867** (4.033)	−9.175** (3.868)	−3.156 (18.332)	−9.017** (4.218)	−8.885** (4.313)
Rel. dependency (young)	0.825 (1.687)	1.801 (1.465)	8.517 (9.676)	0.847 (1.737)	1.391 (1.427)
Rel. dependency (old)	0.522 (1.025)	0.871 (0.906)	1.781 (1.621)	1.107 (1.441)	1.093 (1.224)
Financial deepening	−0.014 (0.009)	−0.018** (0.009)	−0.021 (0.013)	0.017 (0.015)	0.016 (0.015)
ToT volatility	−0.047 (0.044)	−0.030 (0.042)	−0.284*** (0.091)	−0.053 (0.041)	−0.037 (0.040)
Avg. GDP growth	0.423*** (0.111)	0.381*** (0.111)	0.964*** (0.284)	0.400*** (0.106)	0.384*** (0.107)
Openness ratio	−0.039*** (0.007)	−0.036*** (0.007)	−0.050*** (0.018)	−0.040*** (0.012)	−0.040*** (0.010)
Capital controls	2.336** (0.911)	2.297** (0.926)	−1.367 (1.630)	2.995*** (1.150)	3.367*** (1.004)
Oil exporter dummy	−2.608 (1.677)	−3.150*** (1.208)		−2.010 (1.585)	−2.833*** (0.996)
Constant	−0.080 (2.634)	−1.625 (2.355)	−1.102 (18.758)	−0.027 (2.777)	−1.034 (2.331)
Time dummies	Yes	Yes	Yes	Yes	Yes
Observations	295	266	65	230	201
R ²	0.347	0.388	0.889	0.402	0.480
Adjusted R ²	0.302	0.340	0.845	0.348	0.425

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 13: Gross Outflows OLS with time effects

	<i>Dependent variable:</i>				
	Outflows				
	Full	Full excl. Africa	Industrial	Developing	Dev. excl. Africa
	(1)	(2)	(3)	(4)	(5)
Govt. budget balance	0.187* (0.112)	0.239* (0.124)	0.268 (0.421)	0.253* (0.135)	0.295* (0.164)
NFA to GDP ratio	0.047 (1.532)	−0.285 (1.635)	−13.009*** (3.568)	1.213** (0.540)	1.067** (0.464)
Relative income	−7.001 (7.899)	−10.390 (8.001)	−68.235 (120.205)	−14.109** (5.504)	−16.329*** (5.936)
Relative income squared	22.279*** (7.548)	24.618*** (7.558)	71.529 (73.763)	37.585*** (6.635)	38.455*** (6.760)
Rel. dependency (young)	−1.724 (2.171)	−1.982 (2.267)	−3.226 (40.324)	−0.795 (2.351)	−0.765 (2.307)
Rel. dependency (old)	−1.892 (1.771)	−1.207 (1.675)	−0.436 (5.056)	0.426 (2.118)	1.100 (2.078)
Financial deepening	0.009 (0.019)	0.008 (0.020)	0.107* (0.061)	0.013 (0.028)	0.009 (0.032)
ToT volatility	0.024 (0.078)	0.025 (0.078)	0.721*** (0.275)	−0.035 (0.073)	−0.036 (0.073)
Avg. GDP growth	0.040 (0.234)	0.060 (0.246)	−0.926 (1.218)	0.081 (0.222)	0.087 (0.237)
Openness ratio	0.077*** (0.015)	0.083*** (0.016)	0.291*** (0.098)	0.051*** (0.017)	0.058*** (0.020)
Capital controls	−3.176 (2.857)	−4.050 (3.164)	1.609 (5.543)	−1.535 (3.056)	−2.248 (3.360)
Oil exporter dummy	0.664 (2.041)	3.393* (2.018)		1.539 (1.837)	3.120 (2.008)
Constant	−0.948 (3.256)	−0.797 (3.316)	−12.505 (74.200)	1.653 (2.981)	1.522 (2.823)
Time dummies	Yes	Yes	Yes	Yes	Yes
Observations	295	266	65	230	201
R ²	0.443	0.457	0.674	0.566	0.575
Adjusted R ²	0.404	0.415	0.546	0.526	0.530

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 14: Gross Inflows OLS with time effects, with a lagged capital controls variable added

	<i>Dependent variable:</i>				
	Full	Full excl. Africa	Inflows Industrial	Developing	Dev. excl.
	(1)	(2)	(3)	(4)	(5)
Govt. budget balance	−0.214** (0.102)	−0.251*** (0.082)	−0.411*** (0.118)	−0.148 (0.121)	−0.195 (0.088)
NFA to GDP ratio	−0.772 (0.914)	−0.788 (1.010)	−4.962*** (0.716)	0.098 (0.330)	0.130 (0.338)
Relative income	7.850 (4.973)	8.240* (4.683)	4.128 (31.689)	1.109 (5.039)	1.113 (4.816)
Relative income squared	−8.249* (4.227)	−8.385** (4.020)	−5.154 (19.961)	−9.508** (4.468)	−8.93 (4.593)
Rel. dependency (young)	0.784 (1.740)	1.704 (1.529)	8.242 (10.701)	0.954 (1.775)	1.540 (1.488)
Rel. dependency (old)	0.172 (1.074)	0.501 (0.941)	0.998 (1.586)	1.069 (1.441)	1.092 (1.238)
Financial deepening	−0.012 (0.009)	−0.015* (0.009)	−0.021 (0.014)	0.018 (0.016)	0.018 (0.015)
ToT volatility	−0.055 (0.047)	−0.039 (0.045)	−0.261** (0.105)	−0.062 (0.044)	−0.04 (0.042)
Avg. GDP growth	0.411*** (0.116)	0.364*** (0.116)	0.827*** (0.303)	0.389*** (0.111)	0.367*** (0.112)
Openness ratio	−0.040*** (0.008)	−0.037*** (0.007)	−0.052** (0.021)	−0.038*** (0.012)	−0.039 (0.010)
Capital controls	3.586** (1.501)	2.964* (1.583)	−1.240 (3.558)	4.302*** (1.503)	3.978* (1.520)
Capital controls in previous period	−1.480 (1.433)	−0.750 (1.485)	−0.628 (3.048)	−1.727 (1.562)	−0.84 (1.531)
Oil exporter dummy	−2.117 (1.741)	−2.600** (1.184)		−1.524 (1.613)	−2.217 (0.969)
Constant	0.115 (2.696)	−1.251 (2.458)	−2.547 (19.821)	0.082 (2.841)	−0.88 (2.474)
Time dummies	Yes	Yes	Yes	Yes	Yes
Observations	279	250	58	221	192
R ²	0.338	0.378	0.897	0.398	0.472
Adjusted R ²	0.290 ²⁶	0.327	0.850	0.341	0.414

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 15: Variance decomposition: between and within components (updated sample).

Variable	Industrial countries		Developing countries	
	Between	Within	Between	Within
Current account / GDP	47.2	52.8	40.7	59.3
Govt. budget balance	55.0	45.0	35.1	64.9
Relative income	78.1	21.9	84.8	15.2
Relative income ²	78.8	21.2	72.9	27.1
Rel. dependency (young)	66.8	33.2	80.0	20.0
Rel. dependency (old)	52.1	47.9	79.7	20.3
Avg. GDP growth	19.2	80.8	17.7	82.3
Std. dev. GDP growth	19.6	80.4	19.5	80.5
ToT volatility	25.7	74.3	19.3	80.7
REER	33.8	66.2	28.4	71.6
Openness	73.8	26.2	88.4	11.6
Financial deepening	71.7	28.3	26.6	73.4
National saving / GDP	91.1	8.9	74.3	25.7
Capital account openness	25.9	74.1	54.5	45.5
NFA / GDP	43.6	56.4	43.5	56.5

Notes: Each entry reports the percentage of total variance attributable to between-country (cross-sectional) and within-country (time-series) variation, respectively. The oil exporter dummy is excluded (degenerate decomposition).

Table 16: OLS, updated sample (no intercept).

	(1) Full	(2) Full excl. Africa	(3) Industrial	(4) Developing	(5) Dev. excl. Africa
Govt. budget balance	0.359*** (0.059)	0.343*** (0.063)	0.320* (0.139)	0.405*** (0.076)	0.376*** (0.070)
NFA to GDP	1.427*** (0.277)	0.953*** (0.271)	1.844** (0.684)	0.680* (0.321)	0.190 (0.292)
Relative income	-1.280 (3.918)	-11.582** (3.906)	3.552 (35.476)	0.587 (4.940)	-17.461*** (4.336)
Relative income ²	4.533 (3.445)	13.919*** (3.385)	4.123 (21.107)	12.694* (5.398)	31.254*** (4.807)
Rel. dependency (young)	-3.318*** (0.856)	0.110 (1.287)	-5.865 (6.936)	-3.624*** (1.018)	0.502 (1.292)
Rel. dependency (old)	-1.459 (0.759)	-0.146 (0.755)	1.870 (1.718)	-2.423* (1.212)	0.188 (1.030)
Avg. GDP growth	-0.231* (0.091)	-0.260** (0.094)	-0.872* (0.395)	-0.093 (0.104)	-0.192* (0.093)
GDP growth std. dev.	-0.016 (0.102)	0.071 (0.116)	-0.338 (0.409)	-0.054 (0.117)	0.135 (0.117)
ToT volatility	-0.024 (0.033)	-0.037 (0.034)	0.137 (0.169)	0.069 (0.038)	-0.047 (0.034)
REER	0.863* (0.357)	-0.066 (0.430)	-2.186 (3.553)	0.921* (0.454)	-0.082 (0.456)
Openness	0.019*** (0.005)	0.024*** (0.006)	0.058** (0.020)	0.001 (0.007)	0.004 (0.007)
Financial deepening	-0.001 (0.0003)	0.026*** (0.008)	0.007 (0.014)	-0.001* (0.0003)	0.031** (0.010)
Capital account openness	-0.872 (0.694)	-1.801* (0.760)	2.897 (2.037)	-0.602 (0.812)	-1.651* (0.815)
Oil exporter dummy	2.478* (1.057)	5.102*** (1.324)	—	1.521 (1.161)	5.291*** (1.291)
Observations	502	357	69	385	288
R^2	0.401	0.406	0.718	0.431	0.458
Adjusted R^2	0.384	0.381	0.652	0.409	0.430

Notes: Dep. var.: current account to GDP ratio. OLS without intercept. Standard errors in parentheses. The oil exporter dummy is collinear with other regressors in the industrial countries sample and is omitted. $p < 0.10$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

Table 17: Two-way fixed effects (country and time), updated sample.

	(1) Full	(2) Full excl. Africa	(3) Industrial	(4) Developing	(5) Dev. excl. Africa
Govt. budget balance	0.340*** (0.061)	0.269*** (0.067)	0.827*** (0.214)	0.329*** (0.065)	0.265*** (0.073)
NFA to GDP	0.643* (0.269)	0.226 (0.244)	1.114 (0.793)	0.508 (0.293)	0.029 (0.262)
Relative income	22.968** (7.794)	9.412 (7.648)	169.910 (103.703)	28.175*** (8.393)	14.158 (8.226)
Relative income ²	-1.210 (6.462)	10.214 (6.131)	-99.159 (62.533)	-5.460 (7.385)	5.363 (6.846)
Rel. dependency (young)	-0.840 (1.823)	1.120 (2.271)	0.588 (20.008)	-1.646 (1.898)	-0.439 (2.484)
Rel. dependency (old)	-1.549 (1.620)	3.126 (1.604)	-1.625 (3.747)	-3.092 (1.997)	3.439 (1.986)
Financial deepening	-0.0001 (0.0003)	-0.026 (0.015)	-0.026 (0.036)	-0.0001 (0.0003)	-0.027 (0.018)
Capital account openness	-0.830 (0.928)	-0.217 (0.893)	3.356 (3.302)	-1.334 (0.985)	-0.987 (0.936)
Country FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Observations	562	389	69	493	320
R^2	0.161	0.187	0.522	0.150	0.174
Adjusted R^2	-0.010	0.023	0.244	-0.030	-0.002

Notes: Dep. var.: current account to GDP ratio. Two-way within estimator with country and time fixed effects. Unbalanced panel. Standard errors in parentheses. $p < 0.10$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

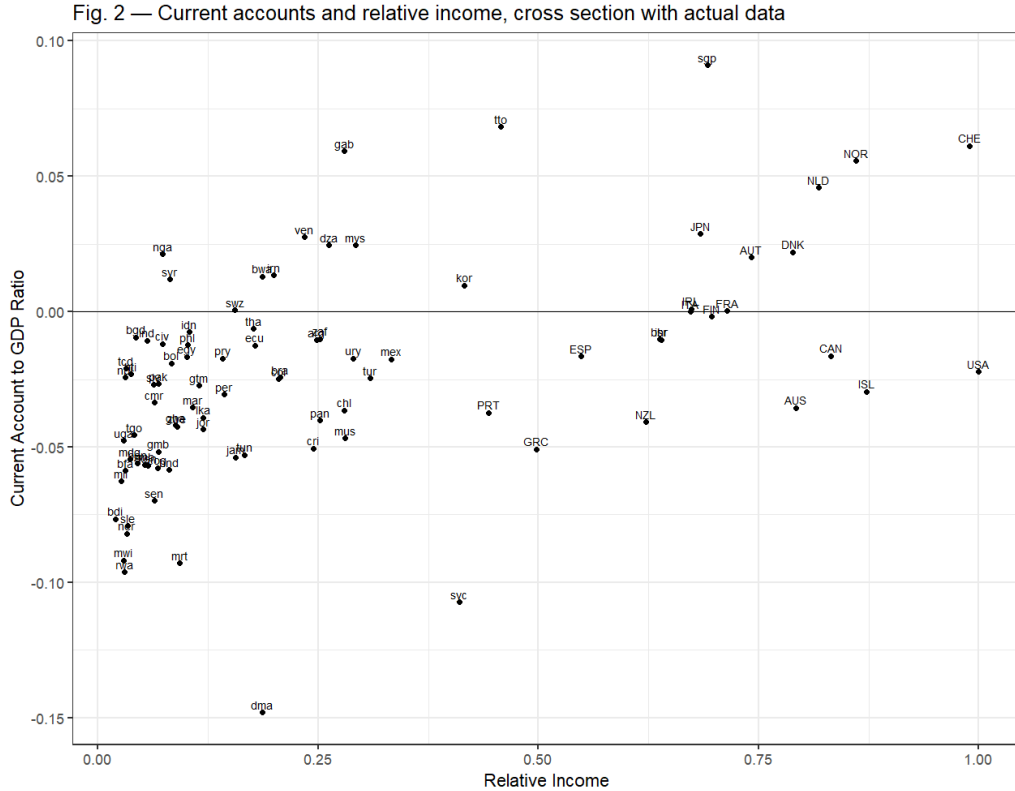


Figure 4: illustrates that higher relative income levels generally correspond with stronger, positive current account positions among wealthier industrial economies like Switzerland and Norway. In contrast, low-income nations exhibit wide vertical dispersion, showing substantial current account deficits despite sharing similarly low relative income levels.

Table 18: OLS, annual data 1971–present, with time effects.

	(1) Full	(2) Full excl. Africa	(3) Industrial	(4) Developing	(5) Dev. excl. Africa
Govt. budget balance	0.197*** (0.042)	0.105*** (0.027)	0.212*** (0.074)	0.225*** (0.051)	0.133*** (0.034)
NFA to GDP	0.642** (0.316)	0.359 (0.282)	0.028 (0.413)	0.773 (0.525)	0.086 (0.432)
Relative income	2.556 (1.620)	−1.011 (1.380)	33.016** (15.866)	1.606 (2.277)	−3.255 (2.058)
Relative income ²	−1.824 (1.360)	1.313 (1.179)	−18.565** (9.119)	1.312 (2.918)	6.585** (2.670)
Rel. dependency (young)	−0.302 (0.385)	0.077 (0.533)	−4.389 (4.210)	−0.292 (0.456)	0.076 (0.583)
Rel. dependency (old)	−0.335 (0.306)	0.024 (0.258)	0.007 (0.816)	−0.288 (0.404)	−0.064 (0.355)
Financial deepening	0.009** (0.003)	0.006** (0.003)	0.007 (0.006)	0.011* (0.006)	0.012** (0.006)
ToT volatility	0.005 (0.004)	0.002 (0.004)	0.013 (0.012)	0.006 (0.004)	0.002 (0.004)
Avg. GDP growth	−0.161*** (0.040)	−0.184*** (0.042)	−0.308*** (0.105)	−0.148*** (0.041)	−0.164*** (0.043)
Openness	−0.002 (0.005)	0.003 (0.004)	0.016 (0.010)	−0.006 (0.006)	−0.004 (0.006)
Capital controls	−0.655** (0.263)	−0.782*** (0.281)	0.941 (0.975)	−0.737** (0.328)	−0.830** (0.340)
Oil exporter dummy	0.026 (0.913)	0.843 (0.844)	—	0.161 (0.899)	1.163 (0.826)
Lagged CA/GDP	0.711*** (0.033)	0.749*** (0.031)	0.762*** (0.080)	0.686*** (0.035)	0.705*** (0.035)
Lagged $\Delta \log$ REER	−3.041*** (0.697)	−2.859*** (0.667)	−7.907*** (2.879)	−2.685*** (0.754)	−2.521*** (0.703)
Constant	2.091* (1.238)	2.745** (1.285)	−12.942 (9.940)	2.840* (1.674)	4.079** (1.763)
Year dummies	Yes	Yes	Yes	Yes	Yes
Observations	1,837	1,391	305	1,532	1,086
R^2	0.661	0.704	0.871	0.624	0.660
Adjusted R^2	0.649	0.691	0.839	0.609	0.640

Notes: Dep. var.: current account to GDP ratio. OLS with year dummies on annual data. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.